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Finalist—2018 MSOM Data Driven Research Challenge

Go Wide or Go Deep? Assortment Strategy and Order Fulfillment in Online Retail

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Abstract. *Problem definition:* Expansions in product assortment by online retailers often engender operational challenges. In undertaking such expansions, retailers exercise a strategic choice between expanding assortment width or depth. Our understanding of how this choice affects the order fulfillment process is limited. Thus, we examine the impact of these dimensions of assortment strategy on order delivery timeliness. *Academic/practical relevance:* Order delivery timeliness is a critical measure of operational success in online retail. We contribute to theory and practice by adopting a multidimensional perspective of retailer assortment strategy and studying the relative impact of assortment width and depth on order delivery timeliness. *Methodology:* Employing a data set comprising more than 200 million orders, we study the effects of assortment strategy on delivery timeliness using an instrumental variable approach. We then utilize a two-stage model to estimate the impact of delivery performance on sales. Further, we employ a matched difference-in-differences and a novel Bayesian structural time-series model to confirm this relationship. *Results:* We find that assortment width has a greater negative impact on order delivery timeliness compared with assortment depth. A one-standard-deviation increase in assortment width increases average delivery times by 0.55 days. Further, we find this effect to be positively moderated (i.e., worsened) by the average size of orders and to be negatively moderated (i.e., improved) by the logistic service provider's (LSP) experience. Finally, a one-day increase in delivery times for 10% of the orders results in a 2.7% reduction in sales. *Managerial implications:* Our findings suggest that online retailers focused on ensuring timely deliveries should be wary of widening product assortments, especially when facing larger average order sizes. We also find that experienced logistic service providers can help mitigate the dilatory effects of assortment width expansions. However, the benefits of experienced LSPs are limited for retailers deepening their assortments.

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Keywords: assortment planning • order fulfillment • empirical operations • online retail

1. Introduction

Advancements in digital technology have opened new markets and enabled the steady growth of electronic commerce to a trillion-dollar global industry. Online retail sales in the United States during 2017 amounted to \$453.5 billion, a 16% year-on-year growth from 2016 (U.S. Department of Commerce 2018). Online retail is characterized by a larger product assortment offered to consumers relative to traditional brick-and-mortar retail channels. For instance, Walmart's online retail arm offers six times the number of products compared with its physical supercenter stores (Brynjolfsson et al. 2003). Although these larger product assortments offered by e-commerce platforms create a consumer

surplus (Brynjolfsson et al. 2003), they raise operational challenges that can potentially negatively impact service quality. In this study, we investigate how an online retailer's *product assortment strategy* affects *order delivery timeliness*.

Product assortment, because of the profound impact it can have on sales and profitability (Kekre and Srinivasan 1990, Lancaster 1990, Bayus and Putsis 1999, Gourville and Soman 2005, Xia and Rajagopalan 2009), is an important strategic decision made by retailers. In traditional offline retail, this strategic choice is made while balancing various constraints (Kök et al. 2008), such as limited budgets, shelf space, etc. In the online setting, where products are directly delivered to

customers' doorsteps, the consideration of delivery constraints while choosing the product assortment strategy assumes special significance as poor delivery performance can adversely affect customers' service experience (Fisher et al. 1999, Wan et al. 2012). High-quality delivery services (e.g., same-day delivery) are viewed as a competitive differentiator by online retailers and have been understood to improve customer satisfaction, boost retention, and ultimately increase profits (Cui et al. 2020). Although there is increasing recognition of the importance of order fulfillment performance in online retail, there is little empirical evidence on the effect of assortment strategy on this crucial measure of service quality.

We note several reasons why addressing this gap is important. One, Gallino and Moreno (2018) note that order fulfillment in an online setting is fundamentally different from an offline setting. In offline retail, transactions are generally fulfilled in-store. In online retail, however, transactions are generally fulfilled by third-party logistic service providers (LSPs) who directly ship products to the customer from a central warehouse. A downside of this approach is that retailers who devise product assortment strategies (Kadiyali et al. 2000, Dukes et al. 2009) can become far removed from the inventory handling challenges associated with different product assortments. Gaining insights on how, and the conditions under which, their product assortment strategies affect order fulfillment, thus, can be valuable for retailers. Two, Brynjolfsson et al. (2003, p. 1581) note that online retailers have "nearly unlimited virtual inventory" and, therefore, can offer a much larger assortment of products to consumers than offline retailers. This larger assortment of products can exacerbate operational problems traditionally faced by offline retailers. Understanding the effect of product assortment strategies on order fulfillment performance, therefore, assumes special significance in the case of online retail.

Three, from a research standpoint, although prior studies have examined the effects of product assortment strategies on several operational factors, such as overhead hours, rework, inventory (Fisher et al. 1999), fill rate (Wan et al. 2012), defect rate (Ton and Raman 2010), and product recalls (Shah et al. 2017), their impact on logistics and order fulfillment, in particular, has not been adequately studied (Ramdas 2003). Because order fulfillment performance serves as an important indicator of a retailer's overall service performance (Lee and Whang 2001), studying it becomes important.

Moreover, in examining the effects of product assortment on operational performance, empirical studies have also generally focused on only one dimension of assortment strategy (i.e., variety) (e.g., Ton and Raman 2010, Wan et al. 2012, Hong et al. 2016). The product assortment strategy of retailers, however, can be distinguished along two dimensions: *assortment depth* and *assortment width* (Kök et al. 2008). Assortment depth

refers to the average number of products offered within a category (Hamilton and Richards 2009 or see Dragan-ska and Jain 2005). Thus, a retailer adopting a deep assortment strategy (such as BestBuy) offers a larger variety of products within the same or a narrow set of categories of products. Assortment width, on the other hand, is defined as the number of unique categories, in which a retailer offers products. Therefore, a retailer adopting a wide assortment strategy (such as Costco) offers products across numerous distinct categories (Cachon and Kök 2007). Following this conceptualization of product assortment, in this study, we investigate the effects of assortment depth and width on order delivery timeliness.

Additionally, the effects of product assortment depth and width on order delivery timeliness, prior literature suggests, can depend on the characteristics of orders faced by online retailers. Acimovic and Graves (2015) and Cachon et al. (2018) highlight how fulfilling larger orders can potentially exacerbate inventory handling challenges. Following this literature, in this study, we also investigate how the effects of product assortment depth and width on order delivery timeliness is influenced by the average size of orders faced by retailers.

Furthermore, the digitization of supply chains has enabled the rapid ongoing shift toward outsourcing of logistic functions to third-party LSPs (see, e.g., Gong et al. 2016). Outsourcing allows online retailers to focus on their core competencies while delegating the responsibility for order fulfillment to specialized logistic firms. Hiring high-quality LSPs assumes special significance in online retail considering the complexity and relative importance of order delivery timeliness as a critical driver of profitability (Rao et al. 2011, Deshpande and Pendem 2022). Prior operations management (OM) literature suggests that delivery timeliness in online retail can also depend on the experience of LSPs employed for order fulfillment. Specifically, Mao et al. (2019) provide evidence to suggest that providers with local area knowledge and prior delivery experience can significantly reduce late deliveries in an on-demand meal delivery platform. Building on this, we investigate how the effects of product assortment depth and width on order delivery timeliness is influenced by LSP experience. Our investigation into the moderating effects of LSP experience is also motivated by recent research that indicates that the value of high-quality logistics (Cui et al. 2020) might be contingent on the product offerings of retailers. Hiring experienced LSPs can be expensive, and there exists little empirical evidence associating LSP experience with favorable outcomes in order fulfillment.

Employing a unique data set provided by Cainiao, Alibaba's affiliated logistics arm, we find that both dimensions of a retailer's assortment strategy, that is, assortment depth and width, adversely affect delivery times. Specifically, a unit increase in assortment width

leads to a 1.7% increase in average delivery times. In comparison, a unit expansion in assortment depth leads to a 0.7% increase in average delivery times. That is, expansions in assortment width have a greater adverse effect on order delivery timeliness than expansions in assortment depth. We further find that this negative effect of assortment width on order delivery timeliness is compounded in retailers experiencing large average order sizes. Finally, we find that hiring experienced LSPs can act as a mitigating lever for merchants expanding their assortment width. However, for retailers that are deepening their assortment, the benefits of working with experienced LSPs are limited.

Moreover, to demonstrate the value of delivery timeliness, we employ a difference-in-differences (DiD) approach that exploits the opening of three new warehouses by Cainiao and confirms the effect of delivery performance on sales by estimating that the addition of these new warehouses resulted in an increase of revenue of 7.3% for the retailers who employ Cainiao as their logistic provider. We also reestimate the above effect using a Bayesian structural time series approach (Brodersen et al. 2015, Schmitt et al. 2018). The novel technique improves upon certain methodological limitations of the DiD approach, and we provide an illustration of its application to the operations literature. Finally, using a two-stage model, we find that the delivery delays induced by a retailer widening its assortment by one category leads to a 1.8% drop in sales across all categories. In summary, this allows us to quantify the eventual downstream impact of a merchant's assortment strategy on sales via the effects on its order fulfillment performance.

2. Literature and Hypothesis Development

An extensive body of research has been dedicated to understanding the relationship between product assortment and sales performance (refer to, e.g., Kekre and Srinivasan 1990, Lancaster 1990, Bayus and Putsis 1999, Gourville and Soman 2005, Xia and Rajagopalan 2009). This stream of literature suggests that a larger product assortment can increase sales as it enables sellers to cater more effectively to the heterogeneous tastes and preferences of consumers (Gourville and Soman 2005). Notwithstanding these gains in sales, a parallel stream of literature identifies that product assortment expansions can impose substantial operational constraints on firms and can adversely affect their performance (MacDuffie et al. 1996, Fisher et al. 1999, Wan et al. 2012). Now, we hypothesize the effects of expansions in product assortment depth and width on order delivery timeliness under different logistical scenarios.

2.1. Assortment Strategy and Delivery Timeliness

Assortment variety can give rise to idiosyncratic inventory handling challenges along a retailer's logistic chain. Inventory storage and handling needs at the warehouse will vary based on the *distinctive* attributes of each product, such as its size, shape, fragility, and perishability. Retailers offering a larger variety in their assortment will need to manage the unique storage and transportation needs of distinctive products. For example, owing to their specialized cold storage and transportation requirements, Reiner et al. (2013) note how logistic execution is particularly challenging when dealing with perishable and dairy products. Azoury and Miyaoka (2013) note similar logistic challenges faced by processed food manufacturers as they manage assortment variety.

Retailers offering a varied assortment will also need to manage the specialized packaging needs of distinctive products. Such retailers will need to carry a range of packaging options of varying materials and sizes, which, in turn, will induce delays in the picking and packing of orders. Canada Post (2020), in its guidebook for online merchants, advises, "If you sell a variety of products, you may have three or four box sizes. If your product range is limited, one or two may suffice. Using the smallest number of packing options that makes sense for your business will save you money on supplies and take some of the guesswork out of packing, saving you time on order fulfillment." Altogether, the diverse storage, transportation, and packaging needs, by increasing training and learning effort, expanding the role of human judgement in selecting and switching between the available options, and increasing the likelihood of errors and rework, will introduce delays into the order fulfillment process. Thus, we posit the following:

Hypothesis 1. *Assortment width and depth are both negatively associated with order delivery timeliness.*

Both expansions in assortment depth and width can engender idiosyncratic inventory handling and packaging demands. However, compared with the addition of products within a category (expansion in depth), the addition of new product categories (expansion in width) can result in the retailers' products becoming more distinctive in terms of their size, shape, fragility, and perishability (Manchanda et al. 1999, Van Herpen and Pieters 2002). Increased distinctiveness of products can engender more diverse inventory handling demands on retailers and introduce more errors and rework in the order fulfillment process. For instance, consider two online retailers selling a dishwashing bar on an e-commerce platform. The first retailer increases the depth of its assortment by adding an organic dishwashing bar, whereas the second retailer increases its width by adding a liquid detergent. The inventory

handling challenges faced by the second retailer will be more diverse than those faced by the first retailer. We posit, therefore, that delays in order delivery will be greater when assortment width is expanded. Formally,

Hypothesis 2. *Assortment width is more strongly negatively associated with delivery timeliness than assortment depth.*

The inventory handling challenges faced by retailers may also be influenced by the substitutability in demand within and across categories. Prior literature, however, indicates that such substitutions may have limited effects on order delays. One, Gaur and Honhon (2006) assert that any substituted demand is expected to transfer to products that are most similar to a customer's preferred product. Thus, although expansions in assortment depth may increase the likelihood of demand substitutions, substituted demand is most likely to fall on similar products because of the increased availability of similar alternatives. Because similar products also tend to have similar handling, transportation, and packaging needs, such substitutions are likely to have little impact on the diversity of inventory handling challenges.

Two, following Gaur and Honhon (2006), we also note that while demand substitutions can also occur across categories as a result of expansions in assortment width, however, they are less likely because of the higher degree of distinctiveness between products in different categories. For example, a customer wanting to buy dishwashing powder is unlikely to buy liquid laundry detergent instead. However, if such cross-category substitutions were to occur, owing to high product distinctiveness, they will continue to add to the handling, transportation, and packaging challenges for retailers.

Three, as observed by Caro et al. (2020), a unique and well-established advantage of online, or omni-channel, retail is the opportunity it offers retailers to leverage the digital trail and rich data generated by customers' store visits and ordering behaviors to more accurately forecast demand (also see, e.g., Huang and Van Mieghem 2014, Ferreira et al. 2016, Martínez-de Albéniz et al. 2020). Such capabilities can enable retailers to manage demand substitutions more effectively. Thus, overall, we expect delays in order fulfillment to derive primarily from product distinctiveness and the diverse inventory storage, transportation, and packaging needs associated with it. Because these diverse needs will be more acute in the case of width expansions, we expect these expansions to be more strongly negatively associated with delivery timeliness.

2.2. Moderating Effect of Order Size

Inventory handling challenges arising from expansions in assortment width can be influenced by the characteristics

of orders faced by retailers. Specifically, prior studies have examined the effects of order sizes on the order fulfillment process. Jasin and Sinha (2015), for example, develop a model to optimize inventory handling in the presence of multi-item orders. Focusing on assemble-to-order computer products, Song (2002) offers analytical techniques to resolve the inventory handling and service quality trade-offs resulting from multi-item orders. It becomes important, therefore, to account for the average size of orders faced by the merchants while examining their order fulfillment performance. Thus, we now hypothesize the moderating effect of the *average size of orders* on the relationship between product assortments and order delivery timeliness.

The average size of consumers' order basket can vary considerably across different retailers. For instance, Bell et al. (1998) demonstrate how consumers with typically larger order sizes tend to visit stores with different average product prices compared with consumers with smaller order sizes. Further, order sizes can also vary across consumers and shopping trips (Bell et al. 2011).

An increase in the average size of consumers' order basket can result in more distinctive products appearing in the basket. For instance, prior literature suggests that variety-seeking impulses (Pessemier 1978), special marketing promotions, and quantity discounts can drive consumers to buy additional products from different categories (see, e.g., Raju 1992). Consumers can also purchase products from different categories due to the natural complementarities between such products. (Russell and Petersen 2000, Chu et al. 2008). Manchanda et al. (1999, p. 95), for example, note "the choice of one category may affect the selection of another category due to the complementary nature (e.g., cake mix and cake frosting) of the two categories. Additionally, products from two categories can also co-occur in a shopping basket because they have similar purchase cycles (e.g., beer and diapers)." Thus, the inventory handling challenges arising from managing the diverse storage, transportation, and packaging needs of distinctive products will be intensified when retailers face larger order sizes. Hence, we hypothesize that,

Hypothesis 3. *Average order size negatively moderates the effects of assortment depth and width on order delivery timeliness.*

2.3. Moderating Effect of Logistic Service Provider Experience

A vast literature on organizational experience and learning highlights the role of task repetition in improving productivity and reliability (e.g., Schilling et al. 2003, Argote and Miron-Spektor 2011). Cumulative experience gained through task repetition has been shown to contribute toward improved manufacturing reliability (Ramdas and Randall 2008), better surgical outcomes

(Kc and Staats 2012), and higher short-term worker productivity (Staats and Gino 2012). These benefits from cumulative experience are accrued through the transfer of learning from prior experience to perform current tasks, through the application of prior learning to adapt and make adjustments to the current task, and through applying prior learning to learn more quickly and effectively (i.e., *learn to learn*) from changes in the current task (Ellis 1965, Narayanan et al. 2009). Focusing specifically on the context of handling product inventory, for example, Argote and Miron-Spektor (2011) suggest that organizational experience gained would depend on the cumulative number of products handled and that experience gained through repetition with one type of product would prepare organizations to handle similar products more effectively.

Following this literature, we note that LSPs accumulate experience by repeatedly fulfilling product orders. Because products from a category are likely to engender similar inventory storage and handling challenges, experienced LSPs will effectively draw upon and apply their past accumulated experience for handling products from similar categories in the future. Such LSPs will also be more effective in leveraging prior learning to adapt and make adjustments to product handling, transportation, and packaging routines when diverse needs arise through the addition of new categories.

In addition to the learning gained through repeated physical handling of inventory, another key source of learning for logistic organizations is customer feedback (Srinivasan and Moorman 2005, Thirumalai and Sinha 2005). Chen et al. (2021, p. 7472) note, “many service businesses monitor reviews closely for feedback as well as detecting service flaws.” Experienced LSPs will have greater exposure to direct customer interactions and more feedback data for analysis and insights. These interactions and feedback will enable experienced LSPs to be better prepared for handling diverse inventory handling challenges. For example, Amazon’s Frustration Free Packaging program channels customer feedback to vendors and LSPs and allows them to save delivery times through a better balancing of diverse inventory handling and customer needs (Pierce 2022). Overall, experienced LSPs, via task repetition or learning from customer feedback, will be more capable of mitigating the dilatory effects of assortment expansions and inventory handling needs that arise from it. Formally,

Hypothesis 4. *The adverse effects of assortment width and depth on order delivery timeliness would be weaker in the presence of experienced LSPs.*

3. Data Overview

We employ data provided for the 2018 MSOM Data Driven Challenge by Cainiao (Alibaba’s logistic arm)

and Tmall.com, a large Chinese business-to-consumer online retail platform. The data spans 527 merchants using Tmall’s platform altogether fulfilling more than 200 million orders across a seven-month period from January 1, 2017, to July 31, 2017. These data have been employed by recent studies to examine various supply chain phenomena in the online retail context (see Bray 2020, Cui et al. 2020, Deshpande and Pendem 2022). The data contain information on the products offered by merchants alongside detailed logistical information associated with the delivery process (e.g., when an order was placed and when it was delivered). From the data on products offered by merchants, we construct our measures for assortment depth and width. We infer the delivery time for orders as the time elapsed from the payment for orders by customers to the time of receipt of the package. The construction of these variables is described below. We also refer the reader to Table A1 in the online supplement for a summary of the key variables used in our econometric models.

3.1. Key Variables

In order to investigate the effect of assortment strategy on order fulfillment performance, we aggregate the orders placed with each merchant pertaining to the products in a particular category (*Category*, the lowest level of product categorization¹ at the weekly level to smoothen out daily variations). We consider two measures of order delivery performance. For those orders without a promised delivery date, we measure delivery timeliness, $DeliveryTime_{ijt}$, as the average time taken by merchant i to deliver the orders in category j during week t from the time of payment. For those orders with a promised delivery date, we measure the delivery performance by the delays in the order fulfillment process from the promised date, $DelayTime_{ijt}$. We define our main independent variables corresponding to the assortment strategy of the retailers, $Depth_{it}$ and $Width_{it}$. The depth of the assortment offered by a merchant is operationalized as the average number of items in each category that the merchant offers, $Depth_{it}$. The assortment width, $Width_{it}$ is measured as the total number of categories offered by the merchant.

As discussed in Sections 2.2 and 2.3, we examine two key moderators: the average size of orders as well as LSP experience. The average size of orders, $AvgOrderSize_{it}$ is measured as the average number of items in each order placed with merchant i during week t . The experience of LSPs employed by a merchant i is measured as the cumulative number of orders (in millions) in category j delivered by them until week t , $LSPExperience_{ijt}$.

In Section 5, we estimate the downstream impact of delivery timeliness on the future sales of the merchant. To do so, we aggregate the sales of a merchant i across

all items in a category j at the weekly level, $Sales_{ijt}$. Further, we aggregate orders delivered each week (as opposed to orders placed each week as before) and utilize the corresponding average measures of delivery timeliness denoted by $DeliveryTime'_{ijt}$ and $DelayTime'_{ijt}$.

3.2. Control Variables

In our analysis, we incorporate several control variables. $\log(Sales_{it})$ controls for the volume of orders handled by merchant i in week t . Certain merchants might strategically trade off delivery timeliness against their product pricing. We account for this by controlling for the average price of products ordered in a particular category from merchant i during week t , $AvgPrice_{ijt}$, which also controls for the variations in sales arising from differences in pricing strategies. LRS_{it} corresponds to the overall logistic review score that a customer views on the web page of merchant i at week t , and this controls for the expected logistic capabilities of a merchant.² LP_{it} controls for the number of logistic providers utilized by a merchant i during week t .

We provide descriptive statistics for all key dependent and independent variables used in our models in Table A2 in the online appendix. In addition, we also provide pairwise correlations of these variables in Table A3. We do not find high correlations precluding estimation issues that may arise because of multicollinearity.

4. Econometric Model and Results

To understand the relationship between assortment strategy and delivery timeliness, we first estimate the following fixed-effect panel regression model that exploits temporal variations in the assortments offered by the merchants on the online retail platform.

$$DeliveryTime_{ijt} = \alpha_1 Depth_{it} + \alpha_2 Width_{it} + CONTROLS_{ijt} + Merchant_i + Category_j + Week_t + \epsilon_{ijt} \quad (1)$$

We primarily measure the delivery timeliness of a retailer by $DeliveryTime_{ijt}$, the average time, aggregated at the weekly level, for a merchant to satisfy the orders placed during week t pertaining to category j . In Equation (1), the signs of the coefficients α_1 and α_2 allow us to test Hypothesis 1. Further, because a change in assortment variety can equivalently be achieved by a change in either assortment width or depth, the relative magnitude and direction of the estimates of coefficients α_1 and α_2 , of $Depth_{it}$ and $Width_{it}$ respectively, allows us to test Hypothesis 2.

We control for $\log(Sales_{ijt})$ and $AvgPrice_{ijt}$ to account for the effects on delivery performance arising from variations in the volume of sales on a weekly basis or from the valuation of the items being delivered. We also

employ LP_{it} and LRS_{it} to control for the logistic capabilities of merchant i during week t . Specifically, they control for the number of logistic providers utilized and the average logistic score respectively. In Equation (1), $CONTROLS_{ijt}$ denotes all these control variables. We also incorporate the following fixed effects in our regression model: (i) $Merchant_i$ accounts for time-invariant unobserved merchant characteristics that drive both the assortment strategy as well as the merchant's e-commerce order delivery performance; (ii) $Category_j$ accounts for unobserved product characteristics that simultaneously affect the delivery timeliness of orders in that category as well as the assortment offered by a merchant; and finally, (iii) $Week_t$ accounts for unobserved temporal shocks and platform-wide variations at the weekly level. We use these fixed effects consistently in this paper.

The results of the ordinary least squares (OLS) estimation of (1) are provided in column (1) of Table 1. Consistent with our expectation, we find the coefficient of product assortment width (α_2) to be positive and significant, whereas the coefficient of product assortment depth (α_1) is insignificant.

However, these coefficients, we note, may be biased as assortment strategy is an endogenous decision by the online retailer that, in practice, is based on several considerations, one of which may potentially be the expected future order delivery performance itself. In other words, our key explanatory variables, $Width_{it}$ and $Depth_{it}$, could be affected by our dependent variable. Furthermore, despite our fixed effects model, which accounts for time-invariant unobservable characteristics (such as the location of the merchant), omitted

Table 1. Impact of Assortment Strategy on Order Delivery Timeliness

Variable	DeliveryTime	
	(1) OLS	(2) 2SLS
Width	0.020*** (0.004)	0.047*** (0.015)
Depth	−0.001 (0.001)	0.020** (0.009)
LP	0.005*** (0.002)	0.006*** (0.002)
AvgPrice	0.000*** (0.000)	0.000*** (0.000)
LRS	−0.105*** (0.019)	−0.103*** (0.020)
$\log(Sales)$	−0.002 (0.002)	−0.005* (0.003)
Adjusted R ²	0.641	0.642

Notes. Robust standard errors in parentheses. Estimated with merchant, category, and week fixed effects. Column (2) estimated with instruments, $N = 64,233$. 2SLS, two-stage least squares.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

unobserved variables that are correlated with both the assortment strategy decisions and the order delivery performance of retailers could introduce endogeneity bias in our model. Therefore, our estimates in column (1) serve only as preliminary indicators of the effect (Kennedy 2008, Tan et al. 2017). We perform the Hausman test and confirm that there is an endogeneity issue with our estimation. Therefore, to address these endogeneity issues, we employ instrumental variables to estimate the relationship of interest via a two-stage least squares (2SLS) model.

4.1. Instrumental Variable Analysis

Any valid instrument needs to satisfy the following three conditions: (i) it has to be correlated with the endogenous variable (instrument relevance); (ii) it has to be exogenous, that is, it has to affect the dependent variable only via the endogenous variable (exclusion restriction); (iii) it should be strong (i.e., high explanatory power to predict the endogenous variable). In addition, as we have two endogenous variables $Width_{it}$ and $Depth_{it}$, we need at least two instrumental variables for estimation. Although in most studies, data limitations render identifying such instrumental variables challenging, we exploit the richness of our data set, which contains information on the assortment strategy and delivery performance across merchants to construct a pair of valid instrumental variables (of a similar flavor to, for example, Cachon et al. 2018), one for each endogenous variable.

We instrument the merchant's assortment width, $Width_{it}$, for a given merchant i during week t , by considering each category k that merchant i offers at least one product in, and then computing the average number of categories offered by all other merchants competing with merchant i in category k , and finally aggregating these averages over all the categories offered by merchant i . We label this instrument $CompetingWidth_{it}$. Competing merchants are selected as it is anticipated that although the assortment offered by competitors is likely to correlate with the assortment offered by the focal merchant, it is also unlikely to affect the delivery performance of the focal merchant. Therefore, $CompetingWidth_{it}$ serves as a good candidate for an instrument satisfying both the relevance and exclusion restriction criteria.

Similarly, we instrument the merchant's assortment depth, $Depth_{it}$, by aggregating the average number of products offered by competing merchants across each individual category offered by merchant i to obtain $CompetingDepth_{it}$. The average assortment depth of competitors is likely to affect the assortment depth offered by the focal merchant because the variety of products offered in specific categories is typically correlated across competitors; however, the assortment decisions of competitors is unlikely to be based on the expected delivery performance of the focal merchant, thereby

again supporting the validity of the second instrument. We empirically test the relevance and strength criteria for our instruments' validity.

The correlations between the two independent variables and the two instruments are significant and positive. The correlation of $CompetingWidth_{it}$ and $Width_{it}$ is 0.86 ($p < 0.05$) and the $CompetingDepth_{it}$ and $Depth_{it}$ is 0.55 ($p < 0.05$). The first-stage results of the 2SLS estimation (provided in Online Appendix B) show that both instruments predict the two endogenous variables significantly. Therefore, both instruments are relevant (i.e., coefficients are significant at $p < 0.01$). We also find that the F -statistic > 10 affirming that our instruments are not weak. In addition, we conduct the conditional F -statistic tests (Stock and Yogo 2005) and confirm that both instruments are strong (F -statistic for the two instruments are 810.6 and 541.9, respectively).

We now employ the two constructed instrumental variables, $CompetingWidth_{it}$ and $CompetingDepth_{it}$, for the two endogenous variables ($Width_{it}$ and $Depth_{it}$) to implement a 2SLS estimation for the model in Equation (1). Results from the second stage of the 2SLS model are presented in column (2) in Table 1. We find the results of the 2SLS model to be consistent with the base OLS model for width and the coefficient of assortment width to be 0.047 ($p < 0.01$), that is, positive and significant. After correcting for bias through 2SLS estimation, however, we find that the coefficient of product assortment depth to also be positive and significant, 0.020 ($p < 0.05$). Overall, these results suggest that assortment width and depth are both negatively associated with order delivery timeliness and support Hypothesis 1. Specifically, we find that an increase of one unit in assortment width is associated with a 0.047-day increase in average delivery times, which corresponds to nearly a 1.7% increase, whereas an increase of one unit in assortment depth is associated with a 0.02-day or a 0.7% increase in average delivery times. These effect sizes are economically meaningful; for instance, a merchant increasing the assortment width by one standard deviation (i.e., 11.9 units) will result, on average, in a 0.55-day (i.e., more than half a day) increase in delivery times across all orders.

To test Hypothesis 2, we compare the relative effects of assortment depth and width on order delivery times. To interpret these relative effects, we compare the coefficients of assortment depth and assortment width from column (2) of Table 1 using a statistical Z -test (Clogg et al. 1995). The estimates of the coefficients α_1 and α_2 in Equation (1) and their associated robust standard errors, and the statistical Z -test comparing the two, indicates that the claim and $\alpha_2 > \alpha_1 > 0$ are statistically significant. Overall, these results support Hypothesis 2, which states that assortment width has a stronger negative association with order delivery timeliness than assortment depth.

4.1.1. Robustness. We conduct several robustness checks for the results obtained. We provide a brief summary here and more detailed explanations of these robustness checks in Online Appendix C.

First, we consider $DelayTime_{ijt}$, an alternative measure of our key dependent variable order delivery timeliness. $DelayTime_{ijt}$ is the average delay on orders with a promised delivery date. Using this dependent variable, we report our results in Table A5. We find that our results remain robust with an alternative measure of our key dependent variable.

Second, we alternatively measure our key explanatory variables—assortment width and depth at a higher level of categorization. In Table A6, we demonstrate that once again, assortment width has a greater negative impact than assortment depth. Further, we note that at this higher level of categorization, the distinctiveness between products is greater. Our analysis, therefore, demonstrates the robustness of our results against different levels of product distinctiveness.

Finally, we construct a second alternative measure for retailers' assortment expansion strategy. Assortment width and depth can be conceptualized as two orthogonal diversification strategies of the retailers. That is, a retailer widening its assortment is essentially diversifying its assortment offering by adding items across unrelated categories. Whereas, a retailer deepening its assortment is diversifying by adding related items belonging to the same or similar categories. Such diversification strategies have been studied in the context of expansions of firms (Raghunathan 1995, Dewan et al. 1998), teams (Marchetti 2020), and products (Van Herpen and Pieters 2002). Drawing upon this vast literature on diversification, we construct an entropy-based measure (Jacquemin and Berry 1979, Palepu 1985) of product assortment expansion. Using this measure, in Table A7, we find our main results to be consistent. Our analysis further supports the notion that delays in order deliveries derive from increases in product distinctiveness.

4.2. Moderating Effect of Average Order Size

To study the moderating effect of *average order size* on the relationship between assortment strategy and delivery performance, we estimate the following interaction model.

$$\begin{aligned} DeliveryTime_{ijt} = & \beta_1 Depth_{it} + \beta_2 Width_{it} + \beta_3 AvgOrderSize_{it} \\ & + \beta_4 Width_{it} \times AvgOrderSize_{it} \\ & + \beta_5 Depth_{it} \times AvgOrderSize_{it} \\ & + CONTROLS_{ijt} + Merchant_i \\ & + Category_j + Week_t + \epsilon_{ijt} \end{aligned} \quad (2)$$

Table 2. Moderating Impact of Average Order Size on Delivery Timeliness

Variable	DeliveryTime
<i>AvgOrderSize</i>	−0.380** (0.190)
<i>Width</i>	0.034** (0.016)
<i>Depth</i>	0.017* (0.009)
<i>Width</i> × <i>AvgOrderSize</i>	0.010*** (0.003)
<i>Depth</i> × <i>AvgOrderSize</i>	0.003*** (0.001)
Adjusted R ²	0.641

Notes. Robust standard errors in parentheses. Estimated with merchant, category, and week fixed effects and all controls. Estimated with instruments, $N = 64,233$.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

$AvgOrderSize_{it}$ is measured as the average number of items in each order basket placed with merchant i during week t . Similar to our previous approach, to address endogeneity, we estimate the effect employing instrumental variable regression using 2SLS. We present our results in Table 2.

We find that the coefficients of both $Width \times AvgOrderSize$ and $Depth \times AvgOrderSize$ are positive and significant (0.010 and 0.003, respectively). These results support Hypothesis 3 and suggest that average size of orders negatively moderates (i.e., worsens) the effects of assortment width and depth on order delivery timeliness.

Overall, the results for the moderating effect of order size indicates that the appearance of more distinctive products in larger order sizes are likely to exacerbate inventory handling challenges resulting from assortment expansion. These results imply retailers experiencing or anticipating larger order sizes, for example, due to ongoing marketing promotions or seasonality, should be cautious with expansions in product assortment, especially for expansions in assortment width.

4.3. Effects of Logistic Service Providers' Experience

To understand the differential moderating impacts of LSP experience on the relationship between assortment strategy and order delivery timeliness, we estimate the following interaction model given by Equation (3).

$$\begin{aligned} DeliveryTime_{ijt} = & \gamma_1 Depth_{it} + \gamma_2 Width_{it} + \gamma_3 LSPEXperience_{ijt} \\ & + \gamma_4 Width_{it} \times LSPEXperience_{ijt} \\ & + \gamma_5 Depth_{it} \times LSPEXperience_{ijt} + CONTROLS_{ijt} \\ & + Merchant_i + Category_j + Week_t + \epsilon_{ijt} \end{aligned} \quad (3)$$

Table 3. Moderating Effect of LSP Experience on Order Delivery Timeliness

Variable	<i>DeliveryTime</i>
<i>Width</i>	0.0642*** (0.0173)
<i>Depth</i>	0.0192** (0.0087)
<i>LSP Experience</i>	0.0278** (0.0097)
<i>Width × LSP Experience</i>	−0.0012*** (0.0004)
<i>Depth × LSP Experience</i>	−0.0000*** (0.0000)
Adjusted R^2	0.538

Note. Robust standard errors in parentheses. Estimated with merchant, category, and week fixed effects and all controls. Estimated with instruments, $N = 60,511$.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

We report our results in Table 3 using an instrumental variable regression as before. We find that the effect sizes³ of the interaction terms (*Width × LSP Experience* and *Depth × LSP Experience*) are negative and significant. Therefore, experienced LSPs do mitigate the adverse impacts of assortment expansions on order delivery timeliness, thereby supporting Hypothesis 4. Further, importantly, comparing the coefficients using the Z-test, we find that experienced LSPs are most beneficial for retailers that are expanding their assortment width, whereas the benefits of experienced LSPs are much more limited for retailers with deeper assortments⁴ (with logistic experience yielding approximately only about 6% of the benefits for a unit depth expansion in relation to a unit width expansion).

Experienced LSPs, these results indicate, are effective in applying the knowledge gained from repeated handling of product categories in prior orders to mitigate delays that may arise because of assortment expansions, particularly width expansions, by retailers.

To summarize, we consistently find that (i) assortment expansions lead to delays in order delivery; (ii) assortment width has a greater negative impact on order delivery timeliness relative to assortment depth; (iii) these negative impacts are compounded in retailers that are experiencing larger orders on average; and finally, (iv) experienced LSPs can help mitigate the negative impacts of assortment width expansions on order delivery performance, whereas the benefits of hiring experienced LSPs are more limited for deep retailers.

5. Delivery Performance and Future Sales

We now evaluate the downstream impacts of order delivery timeliness on merchants' sales performance. Poor order fulfillment performance by retailers in online platforms, an environment where attracting and retaining customers can be challenging because of the easy

availability of multiple competing alternatives and reduced search costs (Brynjolfsson and Smith 2000), can quickly result in customer dissatisfaction and an erosion of trust in retailers' service quality (Heim and Sinha 2001). This reduction in customer satisfaction and trust can adversely impact retailers' profitability (see, for example, Devaraj et al. 2002, Collier and Bienstock 2006).

Although a few studies have examined some negative downstream consequences of delayed order fulfillment in e-commerce, such as increased product returns and decreased repeat purchases (Rao et al. 2011), a systematic evaluation of the monetary cost of lost future sales, associated with delayed deliveries or with reneging a promise made on delivery times, is scarce, primarily because of data limitations as also noted by Fisher et al. (2019), who address a closely related question by quantifying the increase in sales arising as a result of a new warehouse opened by an online retailer selling direct to consumers.

In addition to the direct negative effects on the sales of a category in which a retailer performs poorly (i.e., the focal category), we note that delayed order fulfillment by a poorly performing retailer in categories other than the focal category can have spillover negative effects on the sales of products in the focal category. Such spillover effects can occur because of two reasons. First, certain products or categories, by virtue of their demand patterns or idiosyncratic supply chain requirements, can pose a greater logistical challenge. Aggrieved customers, however, may not be aware of such distinctive and inherent logistical challenges posed by specific products and may therefore wrongly blame merchant capabilities for delayed deliveries. Second, customers, while deliberating to place an order, may evaluate the retailer on not only the focal product of interest but may also take into account the retailer's holistic order fulfillment performance inferred via overall reviews and ratings. Thus, poor order fulfillment by a retailer in unrelated categories can result in negative spillovers on the future sales of the focal category. Although marketing literature has investigated the presence and magnitude of spillover effects in various contexts (see, for example, Janakiraman et al. 2006, Roehm and Tybout 2006, Anderson and Renault 2013), to the best of our knowledge, spillover consequences of poor operational performance in one particular product or a product category to other products in the assortment stocked by the retailer has not been examined.

We consider two measures of delivery performance. For those orders without a promised delivery date, we measure delivery timeliness, $DeliveryTime'_{ijt}$, as the average time taken by merchant i to deliver the orders in category j during week t from the time of payment. For those orders with a promised delivery date, we

measure the delivery performance by the delays in the order fulfillment process from the promised date, $DelayTime'_{ijt}$.

To estimate the sales impact of delivery timeliness via direct and spillover channels, we employ the following dynamic fixed-effects finite distributed lag model regressing the sales in a particular category, $Sales_{ijt}$ on the past delivery performance by the retailer in the same category measured by $DeliveryTime_{ij(t-k)}$ and $DelayTime_{ij(t-k)}$, and across all other categories, $DeliveryTime_{ij^{-1}(t-k)}$ and $DelayTime_{ij^{-1}(t-k)}$, for $1 \leq k \leq p$, adjusting for the fraction of orders for which the retailer did not promise an expected delivery date given by $f_{ij(t-k)}$ and $f_{ij^{-1}(t-k)}$ respectively, along with other appropriate controls and fixed effects at the merchant, category, and weekly levels. We adopt a distributed lag model because it is likely that the effect of delivery delays on future sales persists for more than one week. We choose a finite number of lags corresponding to a two-month lag ($p = 8$).

$$\begin{aligned} \log(Sales_{ijt}) = & \sum_{k=1}^8 \eta_k f_{ij(t-k)} DeliveryTime'_{ij(t-k)} \\ & + \sum_{k=1}^8 \theta_k (1 - f_{ij(t-k)}) DelayTime'_{ij(t-k)} \\ & + \sum_{k=1}^8 \rho_k f_{ij^{-1}(t-k)} DeliveryTime'_{ij^{-1}(t-k)} \\ & + \sum_{k=1}^8 \sigma_k (1 - f_{ij^{-1}(t-k)}) DelayTime'_{ij^{-1}(t-k)} \\ & + CONTROLS_{ijt} + Merchant_i + Category_j \\ & + Week_t + \varepsilon_{ijt} \end{aligned} \quad (4)$$

We observe that the fraction of orders that are provided a promised delivery date is typically a decision of the retailer or the result of decisions that the retailer can influence. We therefore adopt a two-stage model with the following first stage logit selection equation to predict the fraction of orders that a merchant shall offer a promised delivery date in a particular category during a particular week:

$$\begin{aligned} \log\left(\frac{f_{ijt}}{1 - f_{ijt}}\right) \\ = \zeta_1 AvgPrice'_{ijt} + \zeta_2 LRS_{ijt} + \zeta_3 Width_{it} + \zeta_4 Depth_{it} \\ + Merchant_i + Category_j + Week_t + \varepsilon_{ijt} \end{aligned} \quad (5)$$

In Equation (5), we control for the average price of products delivered in a particular category from merchant i during week t using $AvgPrice'_{ijt}$ alongside controlling for the merchants' assortment and their logistic capabilities.

We provide the results of the selection Equation (5) in Online Appendix D. Subsequently, in the second stage, we substitute the fractions f_{ijt} and $f_{ij^{-1}t}$ with the fitted estimates of the fractions from the selection equation, $\hat{f}_{ijt}$ and $\hat{f}_{ij^{-1}t}$, and reestimate Equation (4). Further, to control for omitted variables such as assortment strategy of the retailer itself, which will affect both the delivery timeliness of the merchant (as in fact shown in Section 4) as well as potentially affect sales, we incorporate a week-merchant interaction fixed effect, which controls for all merchant level weekly variations.

In Equation (4), the cumulative impact of a retailer's delivery timeliness in a particular category on its future sales can be obtained by summing the coefficients, $\sum_{k=1}^8 \eta_k$ and $\sum_{k=1}^8 \theta_k$. The sum of the coefficients $\sum_{k=1}^8 \rho_k$ and $\sum_{k=1}^8 \sigma_k$ will let us identify the presence of spillovers whereby delivery performance in other categories influences the sales of the focal category. The results of the two-stage estimation of (4) along with robust standard errors are also provided in Table A11. of Online Appendix D. We note that the estimates for η_k , θ_k , ρ_k , and σ_k ($1 \leq k \leq 5$) are all negative and significant. For higher lags, expectedly, the significance drops.

We therefore find evidence for the negative impacts of longer delivery times in both the focal and other categories on the future sales of the focal category thereby empirically validating that (i) delivery timeliness is a key influencer of future online retail sales and that (ii) poor order fulfillment performance in other categories has a negative spillover effect on future sales of the focal category. Specifically, we find that a one-day increase in the delivery times for 10% of the orders results in a 2.7% reduction in future sales. Moreover, the delivery delays induced by the addition of one category to the assortment leads to a 1.8% drop in sales across all categories. These negative spillovers provide an economic argument in favour of a logistic review system that highlights variations in order delivery performance of a merchant across product categories.

5.1. Difference-in-Differences

In the preceding discussion, we ignore a source of endogeneity that could potentially arise from online retailers, or their logistic providers as the case may be, planning for the order fulfillment performance in anticipation of future expected sales. This will conservatively bias our estimates from (4). Therefore, in order to address this issue, we perform a robustness check by exploiting the simultaneous opening of three new warehouses by Cainiao, Alibaba's logistic provider, on April 21, 2017, for a difference-in-differences analysis. To adopt a difference-in-differences approach in our context, we firstly identify a treatment (opening of new warehouses) that shall feasibly impact the delivery

timeliness of a certain group of retailers delivering certain categories through Cainiao, *treatment group*, and compare the future sales of the treatment group with an appropriately chosen *control group* before and after the treatment. We observe that Fisher et al. (2019) similarly exploit a quasi-natural experiment of the opening of a new distribution center by an online retailer to identify that the opening of the new distribution center contributed to approximately a 4% increase in revenue on average.

On April 21, 2017, corresponding to week 15 in our horizon, Cainiao opened three new warehouses; from our data, we are able to identify on a weekly basis whether a retailer delivered a particular category of products via Cainiao or not. We conservatively include a merchant-category pair in the treatment group even if only a subset of products in a category was delivered through Cainiao. The treatment is expected to have a positive effect on the delivery timeliness of our treatment group. We initially choose as our control group all other merchant-category pairs that are not delivered through Cainiao. There are 260 merchants in our treatment group and 265 merchants in our control group. Alternately, we choose, more carefully, the control and treatment groups as follows: we match each merchant in the treatment group (delivers through Cainiao) with merchants who do not use Cainiao's logistic service based on the similarity of their offered assortment. Setting a threshold of 80% similarity for matching, we include a firm in the control group only if it is matched to at least one firm in the treatment group. If there are no matched firms for a particular firm in the treatment group, we remove it from our analysis. This leaves us with a treatment group of 225 firms matched to a control group of 203 firms. The treatment shall ostensibly have no effect on the delivery times of orders pertaining to these particular merchants in those categories. A key assumption of the difference-in-differences approach is that before the treatment, the treatment group and the control group should have identical sales trends, the *parallel trends assumption*. We consider a four-week window on either side of the treatment, which occurs at week 15; we statistically test for the parallel trends assumption as follows:

$$\log(\text{Sales}_{ijt+1}) = v_1 \text{TREAT}_{ij} + v_2 w_t + v_3 \text{TREAT}_{ij} \times w_t + v_4 \text{AvgPrice}_{ijt} + \text{Merchant}_i + \text{Category}_j + \varepsilon_{ijt} \quad (6)$$

In Equation (6), the binary variable $\text{TREAT}_{ij} \in \{0, 1\}$ denotes whether a merchant-category pair belongs to the treatment group or the control group; w_t denotes a week variable across the pretreatment period ($1 \leq w_t < 5$) where the treatment is applied during $w_t = 5$. The coefficient v_3 allows us to test for the parallel trends

assumption. If v_3 cannot be statistically differentiated from zero, that is, if the estimate for coefficient v_3 is statistically insignificant in our regression, then we can claim that the parallel trends assumption is satisfied. We provide the results of the estimation of (6) in Table A12 in Online Appendix E (column (1) for our broader treatment-control groups and column (2) for the matched case), and we observe that this is indeed the case. Therefore, accounting for the above controls, the trends in our treatment and control groups are identical before the treatment (the addition of three new warehouses by Cainiao).

Now, we proceed to test whether our treatment induces a posttreatment difference in the trends of our treatment and control groups. In order to test for this, we estimate the following model across the nine-week time span under consideration ($1 \leq w_t \leq 9$):

$$\begin{aligned} \log(\text{Sales}_{ijt+1}) = & \mu_1 \text{POST}_t + \mu_2 \text{TREAT}_{ij} \\ & + \mu_3 \text{TREAT}_{ij} \times \text{POST}_t + \mu_4 \text{AvgPrice}_{ijt} \\ & + \text{Merchant}_i + \text{Category}_j + \varepsilon_{ijt}. \end{aligned} \quad (7)$$

In Equation (7), the binary variable $\text{POST}_t \in \{0, 1\}$ indicates whether we are considering the pretreatment or the posttreatment period. The coefficient μ_1 is the overall effect of the treatment across all merchant-Category pairs in our sample. The coefficient μ_3 is the overall effect of the treatment specifically on the treatment group. We provide the results of the estimation of (7) in Table 4 (column (1) and column (2)), and we observe that μ_3 is positive and significant ($p < 0.05$). The estimates indicate that the opening of three new warehouses and the resultant improvements in delivery performance for retailers employing Cainiao's logistic service led to a 7.3% increase in subsequent sales over the firms in the control group. This justifies our prediction that adding three new warehouses would increase merchant sales by allowing retailers to fulfill their orders more promptly, by adding the ability to

Table 4. Impact of Delivery Timeliness on Future Sales

Variable	(1) Unmatched	(2) Matched sample
POST	−0.0013 (0.0016)	−0.0167 (0.0252)
POST × TREAT	0.0604** (0.0243)	0.0703** (0.0276)
AvgPrice	0.0001*** (0.0000)	0.00005*** (0.0000)
Adjusted R ²	0.630	0.638

Notes. Robust standard errors in parentheses. Estimated with merchant, category, and week fixed effects and all controls. Estimated with instruments, $N = 64,233$.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

offer faster delivery service to customers, and by improving customer satisfaction.

A limitation of the DiD approach is that it assumes the data to be independent and identically distributed ignoring the possibility of serial correlations. Brodersen et al. (2015) generalize the DiD method and describe a Bayesian structural time-series (BSTS) approach to estimate the causal impact of interventions by constructing the counterfactual evolution of the treated time series. The difference between the observed and the counterfactual time series is then identified as the consequence of the intervention. In Online Appendix F, we perform a robustness check using the BSTS approach to confirm the positive relationship between faster order deliveries and sales.

6. Discussion and Conclusion

Our work investigates the impact of product assortment strategies adopted by online retailers on their order delivery performance. We leverage the conceptualization of assortment strategy along two dimensions of depth and width (for example, Kök et al. 2008) to study how an increase in each dimension affects order delivery timeliness. Further, drawing upon prior literature that suggests that customer order characteristics can influence the order fulfillment process (Song 2002, Jasin and Sinha 2015), we investigate the moderating effects of order size on the link between assortment strategy and order fulfillment performance. Finally, we draw upon the growing body of research examining the influence of logistic providers on the order fulfillment process to study the moderating effects of the LSP experience on the assortment strategy-delivery performance link. We provide a summary of the insights stemming from our analysis and results in Table 5.

We find that compared with expansions in assortment depth, expansions in assortment width have a stronger adverse impact on order delivery timeliness. A high level of distinctiveness between products across categories can engender diverse inventory handling challenges that lengthen order delivery times. Although a substantial body of operations and marketing literature (Kekre and Srinivasan 1990, Lancaster 1990, Bayus and Putsis 1999, Gourville and Soman 2005, Xia and Rajagopalan 2009) has been dedicated to understanding the effects of retailer strategies on sales performance, their effects on order fulfillment performance have largely been overlooked. First, our work addresses this important gap. Prior studies highlight that the online environment offers virtually unlimited shelf space to retailers (Brynjolfsson et al. 2003) and demonstrate that product variety enhances sales (e.g., Wan et al. 2012) and customer satisfaction (e.g., Berger and Heath 2007). Thus, although a predominantly positive view of assortment expansions in an online retail context has been previously offered, our findings call for a more

qualified and cautionary approach toward such expansions. Our work specifically underscores the need to account for harmful operational “leakages” in considering the economic impact of product assortment strategies. Fundamentally, our work demonstrates the tight interlinkage between operational and retail assortment strategies.

Second, by demonstrating the differential effects of assortment depth and width on operational performance, our findings further underscore the need to move beyond a monolithic view of product assortment variety (e.g., Ton and Raman 2010, Wan et al. 2012, Hong et al. 2016) and to adopt a multidimensional perspective for a nuanced and complete understanding of this phenomena (e.g., Kök et al. 2008). Practically, our findings caution online retailers to improve their inventory handling and logistical preparedness specifically before launching expansions in product categories. Thus, our work also contributes to prior OM literature on operational preparedness (e.g., Tomlin 2006, Huang et al. 2016) by highlighting when preparedness assumes special operational significance.

Third, assortment decisions related to expansions in width are typically aimed at meeting the customer’s diverse shopping needs (Manchanda et al. 1999, Hong et al. 2016). As the underlying consideration in these decisions is on sales enhancement, little attention is paid to the distinctiveness between the products being added. Our analysis (see Online Tables A6 and A7), however, reveals that ignoring the level of distinctiveness between products can significantly undermine retailers’ operational performance. As product distinctiveness increases, our entropy-based analysis specifically suggests that so do the order delivery times. Fundamentally, our results highlight the need to move beyond adopting a purely structural perspective (number of categories) to a compositional perspective (type and relation between the categories) in studying the operational effects of assortment expansions. From a practical standpoint, our results indicate that online retailers focused on improving delivery times should consider adding new products to an existing category (smaller increase in product distinctiveness) rather than adding new products in a completely new category (larger increase in product distinctiveness).

Additionally, we find that increases in average customer order sizes can exacerbate the dilatory effects of expansions in assortment width on delivery times. Larger orders, by increasing the likelihood of distinctive products from different categories occurring in the same order basket, can intensify the diverse inventory handling challenges faced by retailers and LSPs. Whereas prior research largely focuses on how inventory should be organized for handling multi-item orders (e.g., Song 2002, Jasin and Sinha 2015), our study contributes to extant literature by highlighting the

Table 5. Summary of Key Insights

Hypothesis and proposition	Result reference	Insights
H1: Expansion in assortment width or depth, by adding distinctive products to the offerings, will create diverse storage, mobility, and packaging needs. These diverse inventory handling needs will increase delays in order fulfillment.	Table 1: The coefficients of <i>Width</i> and <i>Depth</i> are positive and significant. The coefficient of <i>Width</i> is significantly higher than that of <i>Depth</i> .	1. Online retailers should plan for the <i>diverse</i> inventory handling needs that can arise from assortment expansions. Logistic preparedness via investments in inventory handling capabilities should be undertaken more acutely when assortment expansions are made by adding new categories.
H2: Expansion in assortment width, by creating more distinctive products in the offerings compared with assortment depth, will engender higher diversity in inventory handling needs and, therefore, greater delays.		2. Logistic preparedness assumes greater importance for assortment width expansions.
H3: Larger orders faced by retailers undergoing assortment expansions are more likely to contain products drawn from distinctive categories. These product combinations will intensify diverse inventory handling needs and result in greater delays.	Table 2: The coefficients of the interaction terms <i>Width</i> × <i>AvgOrderSize</i> and <i>Depth</i> × <i>AvgOrderSize</i> are positive and significant.	3. Online retailers expanding assortments and anticipating larger orders due to, for example, ongoing promotions and discounts etc., should especially plan for and invest in operational and logistical capabilities for handling diverse inventories.
H4: Experienced logistic providers are better placed to effectively adapt to novel inventory handling challenges engendered by assortment expansions.	Table 3: The coefficients of the interaction terms <i>Width</i> × <i>LSPEXperience</i> and <i>Depth</i> × <i>LSPEXperience</i> are negative and significant.	4. Online retailers expanding assortment width should especially consider hiring experienced LSPs. Experienced LSPs are effective in mitigating the adverse impacts of diverse inventory handling challenges emerging from assortment width expansion. The effectiveness of experienced LSPs may be limited for retailers considering expansion in assortment depth.
Proposition: Delays in order deliveries have a negative impact on sales performance. Such poor order fulfillment performance in other categories has a negative spillover effect on future sales of the focal category.	Online Table A11: The coefficients for delivery timeliness terms are negative and significant.	5. Delays in delivery of orders resulting from expansion in product assortment can lead to a potential loss of future sales for online retailers. 6. To prevent negative spillovers across categories, online retailers and platforms can consider adopting a logistic review system that disaggregates information on past delivery performance at the product category level.

Note. H1, Hypothesis 1; H2, Hypothesis 2; H3, Hypothesis 3; H4, Hypothesis 4.

importance of jointly considering assortment strategy and order characteristics. More fundamentally, our work demonstrates how operational outcomes derive from the interactions between retail assortment strategies and consumer choices. Practically, our findings suggest that online retailers facing larger orders should especially be cautious and well prepared operationally before launching new categories. Complementing prior work on order sizes and fulfillment sizes (e.g., Acimovic and Graves 2015, Chen et al. 2019), our work offers product assortment as a strategic lever to control the negatives effects of larger order sizes.

Further, we find that experienced LSPs can mitigate the negative effects of expansions in assortment on order delivery times. By applying their accumulated experience of handling orders from similar product

categories, experienced LSPs can more effectively tackle the diverse inventory handling challenges raised by assortment expansions. Our findings tie into the growing research on third-party logistics. Outsourcing of logistics to third-party service providers has increased with the advent of e-commerce (Gong et al. 2016). In studying the impact of LSPs on order fulfillment, prior literature has primarily focused on LSP capabilities, such as inventory planning (e.g., Lee et al. 2003, Wagenaar et al. 2021), tracking technologies (e.g., Langer et al. 2007), warehousing and consolidation (e.g., Song et al. 2008), LSP quality (e.g., Özpolat et al. 2013, Deshpande and Pendem 2022), and routing and distribution (e.g., Ding and Kaminsky 2020). Our work complements recent efforts by OM scholars to understand how LSP experience influences order fulfillment (Mao et al. 2019).

That prior experience and task repetition improves operational performance has been discussed in organizational literature (Ramdas and Randall 2008, Argote and Miron-Spektor 2011, Kc and Staats 2012, Staats and Gino 2012). Our work demonstrates how LSP experience, by mitigating the downside of assortment expansions, can allow retailers to better exploit the virtually unlimited shelf space offered by online environments and more confidently seek the economic gains associated with larger product varieties. Our work also demonstrates the importance of leveraging LSP experience specifically for retailers considering expansions in assortment width.

Finally, we examine whether delivery delays in online order fulfillment significantly impact a retailer's future sales and the extent to which order delivery performance mediates the relationship between assortment strategy and sales. Prior studies (Cui et al. 2020) have acknowledged the importance of timely deliveries for the overall competitiveness of online retailers. We confirm that order delivery delays in a particular category negatively impacts sales of the focal category as well as sales of other categories. These negative spillovers provide an economic argument in favour of a logistic review system that calls attention to variations in order delivery performance of a merchant across product categories.

6.1. Limitations and Future Research

Our rich data set has enabled us to address key questions pertaining to the operational effects of assortment strategy on order fulfillment in online retail and its eventual impact on future sales. However, we do note certain limitations with the data and our study. First, it is limited in its time span of seven months. Future studies can investigate the effects of assortment strategy and delivery performance on customer repeat rates, customer retention, and other measures of performance that are harder to estimate with a shorter time span. Further, although we employ instrumental variable regression for estimation, one could use field experiments to more precisely understand the mechanisms underlying the assortment strategy and order fulfillment performance link. In future studies, researchers can also focus on the operational impacts of other elements of product assortment strategy, such as assortment consistency defined as the consistency in the assortment offered to customers (Krishnan et al. 2002), and the timing of changes made to the product assortment. Finally, we also propose that researchers investigate the role played by the warehousing strategy of the retailer or logistical providers in managing the operational consequences of adding assortment variety. For example, researchers can study how the geographic location of warehouses and the distribution of a retailer's product assortment inventory across multiple warehouses affects its order fulfillment performance.

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Endnotes

¹ Two levels of product categories are available in our data set. At the lower level, the degree of distinctiveness between products is smaller. This lower level of categorization is akin to categories typically observed in online retail stores. At the higher level, the degree of distinctiveness between products is larger. Adopting a more conservative approach, we conduct all our analysis at the lower level of categorization (henceforth category). A robustness test with our main analysis conducted at the higher level of categorization is also provided in Table A6 (in Online Appendix C.2.). Our results remain consistent across both specifications.

² The average logistic review score is computed by the online platform (Tmall.com) by aggregating the ratings provided by customers over past fulfilled orders.

³ Note that the coefficients correspond to the timeliness benefits of one million additional orders handled by the logistic providers in the associated category in the past.

⁴ We conduct several robustness tests. For details, please refer to Online Appendix C.4.

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